SMPTE Public Committee Draft

Catena — gRPC Connection Type

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Foreword

SMPTE (the Society of Motion Picture and Television Engineers) is an internationally-recognized standards developing organization. Headquartered and incorporated in the United States of America, SMPTE has members in over 80 countries on six continents. SMPTE's Engineering Documents, including Standards, Recommended Practices, and Engineering Guidelines, are prepared by SMPTE's Technology Committees. Participation in these Committees is open to all with a bona fide interest in their work. SMPTE cooperates closely with other standards-developing organizations, including ISO, IEC and ITU.

SMPTE Engineering Documents are drafted in accordance with the rules given in its Standards Operations Manual. This SMPTE Engineering Document was prepared by Technology Committee TC34CS – Media Systems, Control and Services.

Introduction

This clause is entirely informative and does not form an integral part of this Engineering Document.

This document specifies how to transport the Catena model, as defined in SMPTE ST 2138-10, using gRPC, as illustrated in Figure 1.

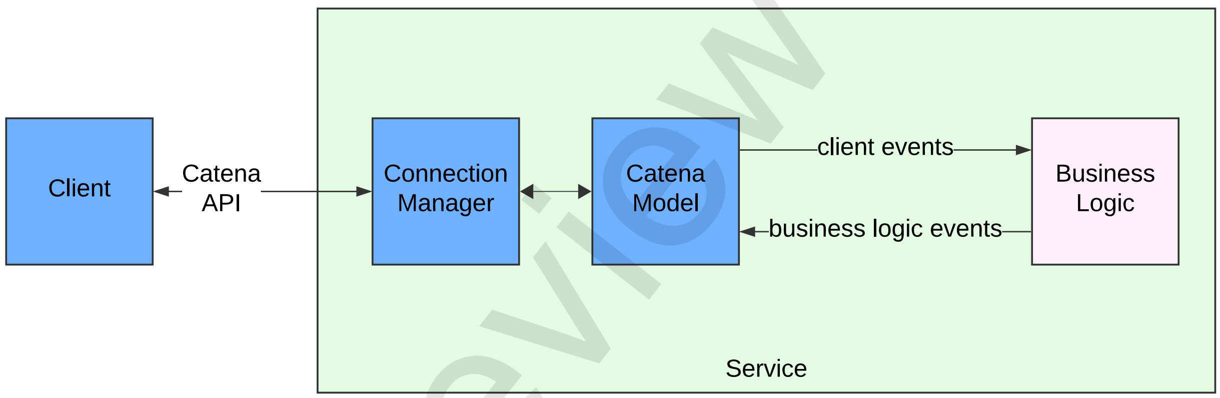

Figure 1 — Relationship of Connection Manager with Catena Model

1 Scope

This document defines the use of a gRPC connection manager with Catena.

2 Conformance

Normative text is text that describes elements of the design that are indispensable or contains the conformance language keywords: "shall", "should", or "may". Informative text is text that is potentially helpful to the user, but not indispensable, and can be removed, changed, or added editorially without affecting interoperability. Informative text does not contain any conformance keywords.

All text in this document is, by default, normative, except: the Introduction, any clause explicitly labeled as "Informative" or individual paragraphs that start with "Note:"

The keywords "shall" and "shall not" indicate requirements strictly to be followed in order to conform to the document and from which no deviation is permitted.

The keywords, "should" and "should not" indicate that, among several possibilities, one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required; or that (in the negative form) a certain possibility or course of action is deprecated but not prohibited.

The keywords "may" and "need not" indicate courses of action permissible within the limits of the document.

The keyword "reserved" indicates a provision that is not defined at this time, shall not be used, and may be defined in the future. The keyword "forbidden" indicates "reserved" and in addition indicates that the provision will never be defined in the future.

A conformant implementation according to this document is one that includes all mandatory provisions ("shall") and, if implemented, all recommended provisions ("should") as described. A conformant implementation need not implement optional provisions ("may") and need not implement them as described.

Unless otherwise specified, the order of precedence of the types of normative information in this document shall be as follows: Normative prose shall be the authoritative definition; tables shall be next; then formal languages; then figures; and then any other language forms.

3 Normative References

The following standard contains provisions that, through reference in this text, constitute provisions of this standard. Dated references require that the specific edition cited shall be used as the reference. Undated citations refer to the edition of the referenced document (including any amendments) current at the date of publication of this document. All standards are subject to revision, and users of this engineering document are encouraged to investigate the possibility of applying the most recent edition of any undated reference.

IETF RFC 3066 - *Tags for the Identification of Languages, H. Alvestrand, Internet Engineering Task Force, 2001*. http://www.ietf.org/rfc/rfc3066.txt

SMPTE ST 2138-19:202x, *Catena – Protocol Objects*

4 Terms, Definitions and Abbreviated Terms

For the purposes of this document, the following terms and definitions apply.

4.1
RPC

Remote Procedure Call

4.2
gRPC

Google's implementation of Remote Procedure Call

5 Precedence

In the event of a conflict between the schema located at additional element at https://github.com/SMPTE/st2138-a.git (in the /interface/schemata/device.yaml file) and other information in this document, the schema shall take precedence. See Annex A for additional information.

6 gRPC Service

6.1 Service Overview

The gRPC Service handled by the Connection Manager shall be comprised of a number of RPCs, summarized in Table 1. Each RPC's sequence diagram is documented here. The UML representations of the objects communicated are presented in the Catena Protocol Objects document SMPTE ST 2138-19:202x.

Request Parameters and Returned Objects may be either unary or streams. Unary means that the Request or Return is complete once a single payload has been communicated. Streaming means that a sequence of objects may be communicated to complete the Request or Return.

NOTE 1 Requests and Returns that are streams are indicated with a trailing ˚ adornment throughout this document.

NOTE 2 Some of the object names are too long to fit easily into a table, so long names have been split by inserting a dash '-' character to wrap the name over multiple lines. Please ignore the dashes.

Table 1 — Catena Service Summary

RPC Name	Request Parameters	Returned Objects	Comments
GetPopulatedSlots	None	SlotList	This is normally the first RPC from a client to a device to determine which slots contain DeviceModels
DeviceRequest	DeviceRequest-Payload	DeviceComponent°	
ExecuteCommand	ExecuteCommand-Payload°	CommandResponse°	
ExternalObjectRequest	ExternalObject-RequestPayload	ExternalObjectPayload°	Opaque object support
BasicParamInfoRequest	BasicParamInfo-RequestPayload	BasicParamInfo-Response°	
SetValue	SetValuePayload	None	
GetValue	GetValuePayload	Value	
MultiSetValue	MultiSetValuePayload	None	e.g., salvo
UpdateSubscriptions	UpdateSubscriptions-Payload	DeviceComponent-.ComponentParam°	
GetParam	GetParamPayload	DeviceComponent.-ComponentParam°	
Connect	ConnectPayload	PushUpdates°	Return stream is open-ended
AddLanguage	AddLanguagePayload	None	
LanguagePackRequest	LanguagePack-RequestPayload	DeviceComponent-.ComponentLanguagePack	
ListLanguages	Slot	LanguageList	
RefreshToken	RefreshTokenPayload	ConnectionStatus	
RevokeAccess	RevokeAccessPayload	RevocationResponse	

Each RPC is defined in https://github.com/SMPTE/st2138-a/blob/main/interface/proto/service.proto and the files that it includes.

6.2 RPC Status Codes

RPCs shall either terminate with a success code, or one of the failure codes as defined in Table 2. For simplicity, the sequence diagrams for each RPC only show the scenario where the returned status code is OK. If a failure occurs, no objects shall be returned by unary responses, and those that might have been returned from streaming responses shall be discarded by the client.

Table 2 — RPC Status Codes

Code name	Value	Comments and Examples
OK	0	RPC completed without error
CANCELED	1	The operation was canceled, by either the client or the device
UNKNOWN	2	Unknown error – there is not sufficient information to use any other status code.
INVALID ARGUMENT	3	Something was wrong with the request. For example, calling the GetValue RPC with an oid that does not map to a parameter would cause this response
TIMEOUT EXCEEDED	4	A time-out, set by the client
NOT FOUND	5	Some requested entity (such as an external object) was not found, probably because the wrong identifier was supplied
ALREADY EXISTS	6	The service was asked to or attempted to create, a file, directory, or other item that already exists.
PERMISSION DENIED	7	The access token's scopes did not include those of the command or parameter accessed.
RESOURCE EXHAUSTED	8	For example, out of disk space
FAILED PRECONDITION	9	The operation was rejected because the service was not in a state necessary for the requested operation to complete
ABORTED	10	For example, the service needed to make transactional calls on other resources and a transaction failed
OUT OF RANGE	11	Attempting to access band 5 of a 4-band eq would create this status code
UNIMPLEMENTED	12	The RPC is not implemented or supported
INTERNAL	13	Used to report unexpected failures by the service.
UNAVAILABLE	14	The service is currently unavailable
DATA LOSS	15	Unrecoverable data loss / corruption
UNAUTHENTICATED	16	The authentication credentials were invalid. For example, the access token was not issued by the registered authorization server

6.3 Remote Procedure Calls (RPCs)

6.3.1 GetPopulatedSlots

This RPC enables clients to query which slots contain device models, as shown in Table 3 and in Figure 2.

Table 3 — GetPopulated Slots RPC Summary

	Transfer Type	Message Definition, file
Request	Unary	Empty, param.proto
Response	Unary	SlotList, device.proto
Reference	SMPTE ST 2138-19:202x Catena – Protocol Objects, Clause 6.2	

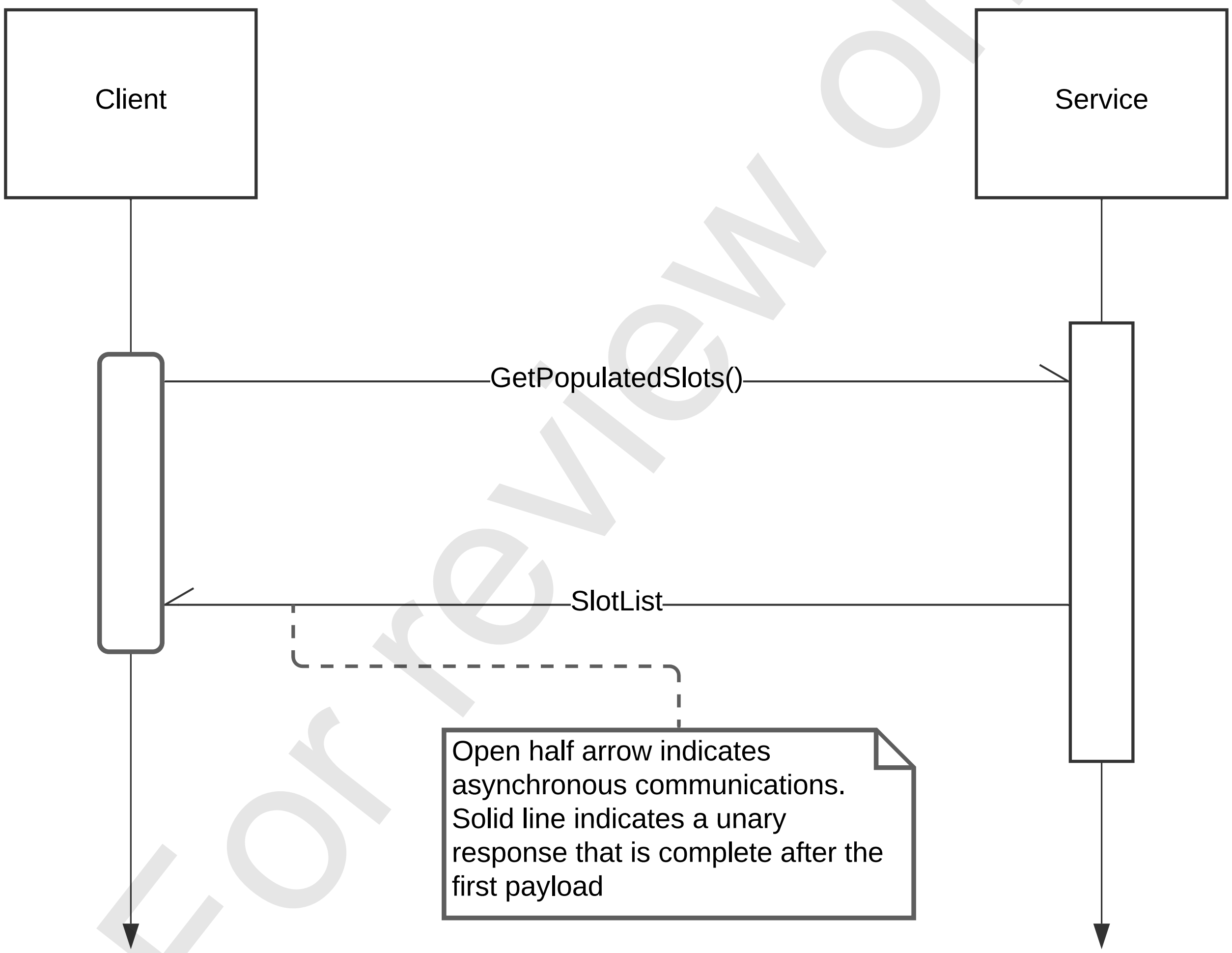

Figure 2 — GetPopulatedSlots Sequence Diagram

6.3.2 DeviceRequest

This RPC, as shown in Table 4 and Figure 3 allows a device to share its device model with connected clients. Those parts of the model to which the authenticated user does not have read access are filtered from the mode.

Table 4 — DeviceRequest RPC

	Transfer Type	Message Definition, file
Request	Unary	DeviceRequestPayload, service.proto
Response	Stream	DeviceComponent, device.proto
Reference	SMPTE ST 2138-19:202x Catena – Protocol Objects Clause 6.3	

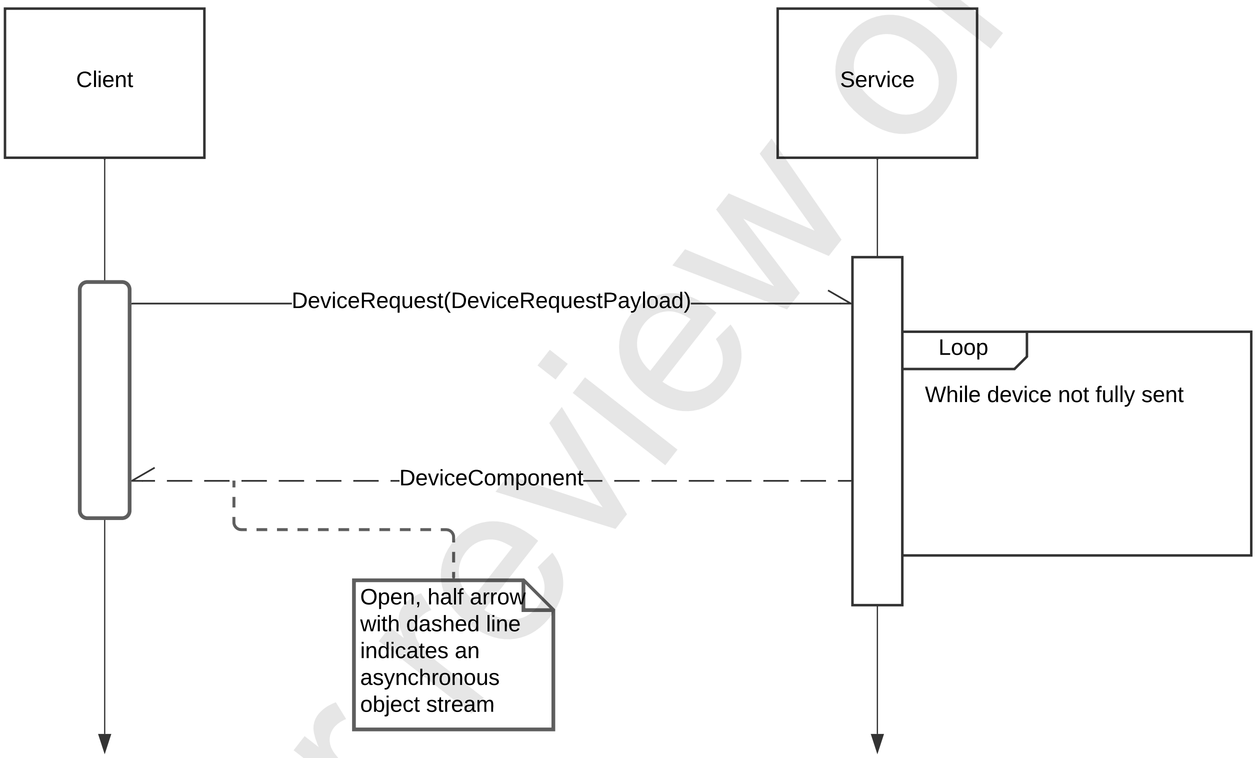

Figure 3 — DeviceRequest Sequence Diagram

6.3.3 ExecuteCommand

This RPC, as shown in Table 5 and Figure 4 enables a client to cause the Commands supported by the attached device to be executed.

Table 5 — Execute Command RPC

	Transfer Type	Message Definition, file
Request	Unary	ExecuteCommandPayload
Response	Stream	CommandResponse
Reference	SMPTE ST 2138-19:202x Catena – Protocol Objects Clause 6.4	

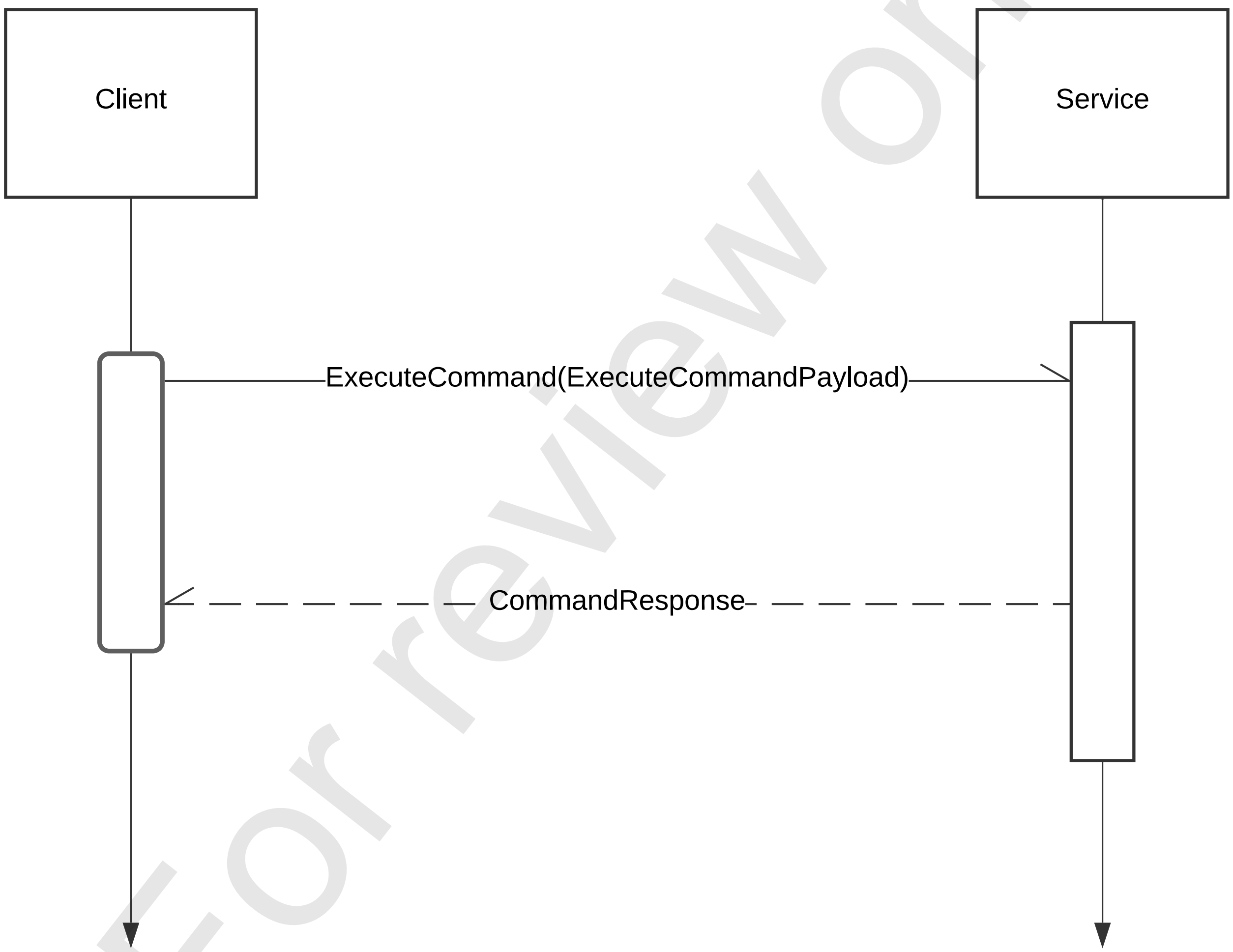

Figure 4 — ExecuteCommand Sequence Diagram

6.3.4 ExternalObjectRequest

This RPC, as illustrated in Table 6 and Figure 5, enables clients to request objects that are not embedded into the device model such as product icons, images, log files and generic data objects.

Table 6 — ExternalObjectRequest RPC

	Transfer Type	Message Definition, file
Request	Unary	ExternalObjectRequestPayload, externalobject.proto
Response	Stream	ExternalObjectPayload, externalobject.proto
Reference	SMPTE ST 2138-19:202x Catena – Protocol Objects Clause 6.5	

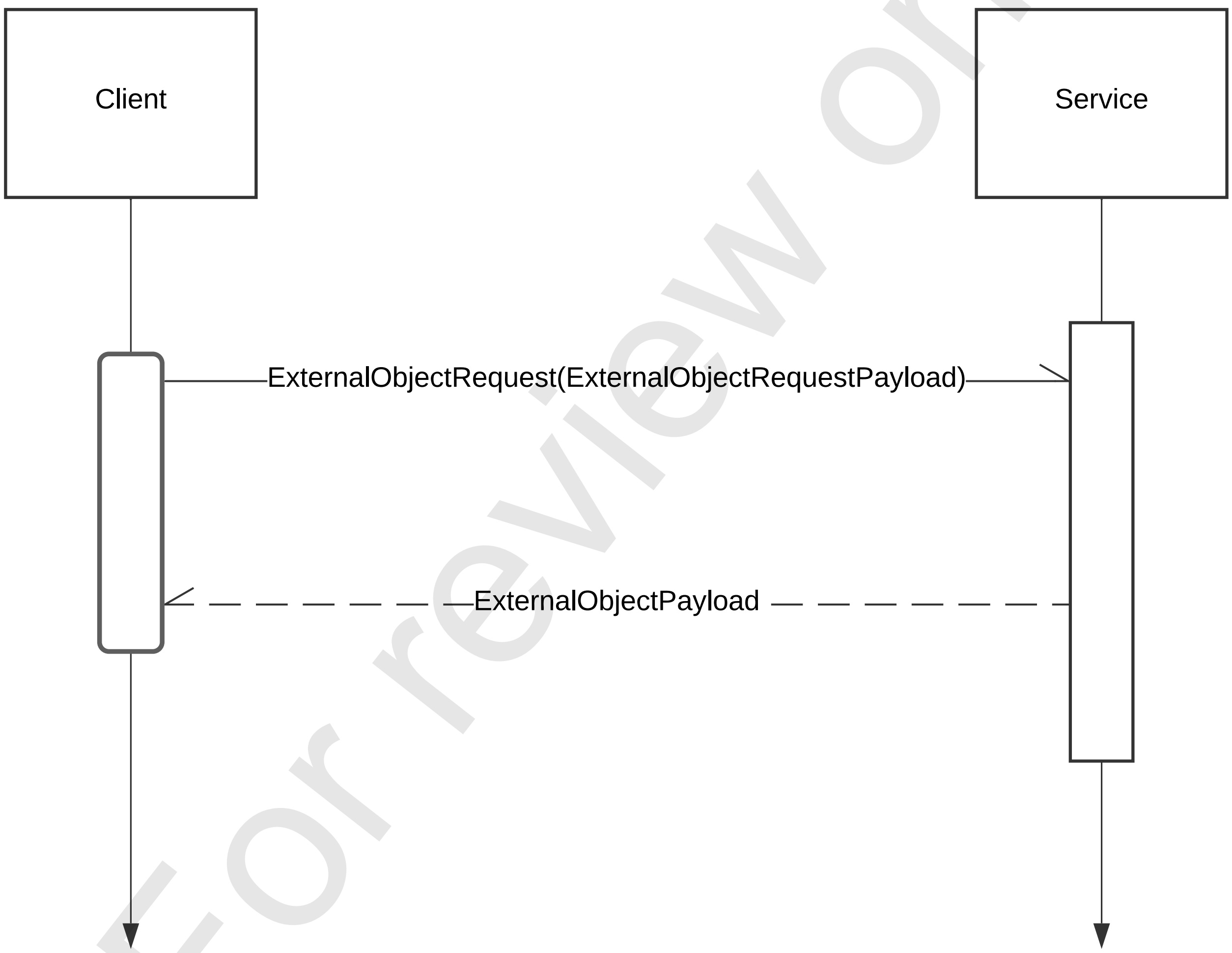

Figure 5 — ExternalObjectRequest Sequence Diagram

6.3.5 BasicParamInfoRequest

This RPC, as illustrated in Table 7 and Figure 6, provides summary information about parameters to clients which is used in managing the parameters to which the client is subscribing.

Table 7 — ParamInfoRequest

	Transfer Type	Message Definition, file
Request	Unary	ParamInfoRequestPayload, param.proto
Response	Stream	ParamInfoResponse, param.proto
Reference	SMPTE ST 2138-19:202x Catena – Protocol Objects Clause 6.7	

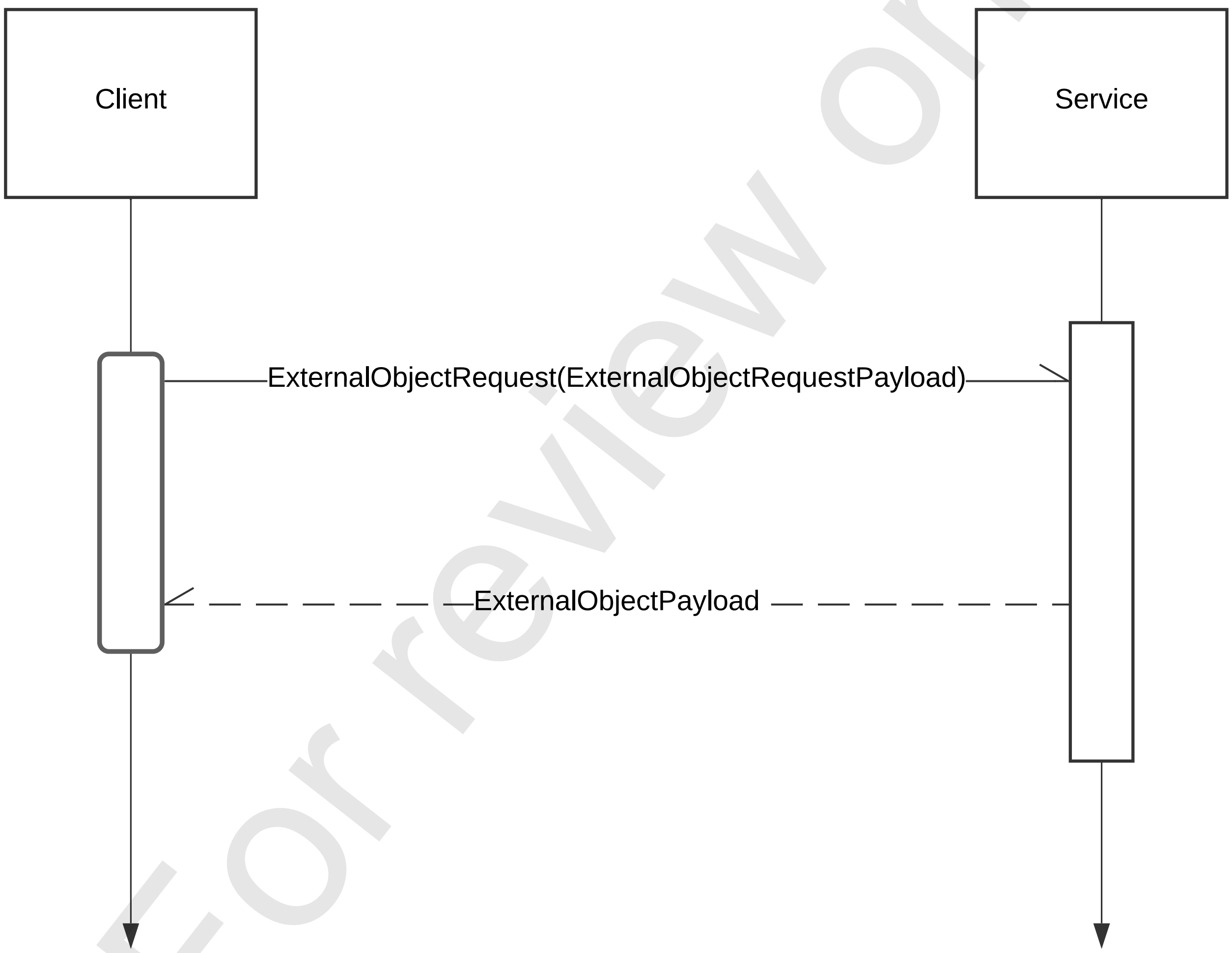

Figure 6 — BasicParamInfoRequest Sequence and Class Diagrams

6.3.6 SingleSetValue and MultiSetValue

Clients use these RPCs (illustrated in Table 8, Figure 7 and Figure 8) to request value changes of the device model's parameters. SetValue shall update only one parameter.

The authenticated user shall have write access to the scopes of all the parameters addressed by this RPC; otherwise, the service shall return an "unauthorized" response. MultiSetValue either succeeds completely or fails completely even if some of the parameters addressed are in-scope, but one is out of scope.

Table 8 — SetValue and MultiSetValue RPC

	Transfer Type	Message Definition, file
Request	Unary	SetValuePayload or MultiSetValuePayload, param.proto
Response	Unary	Empty, param.proto
Reference	None in common with REST	

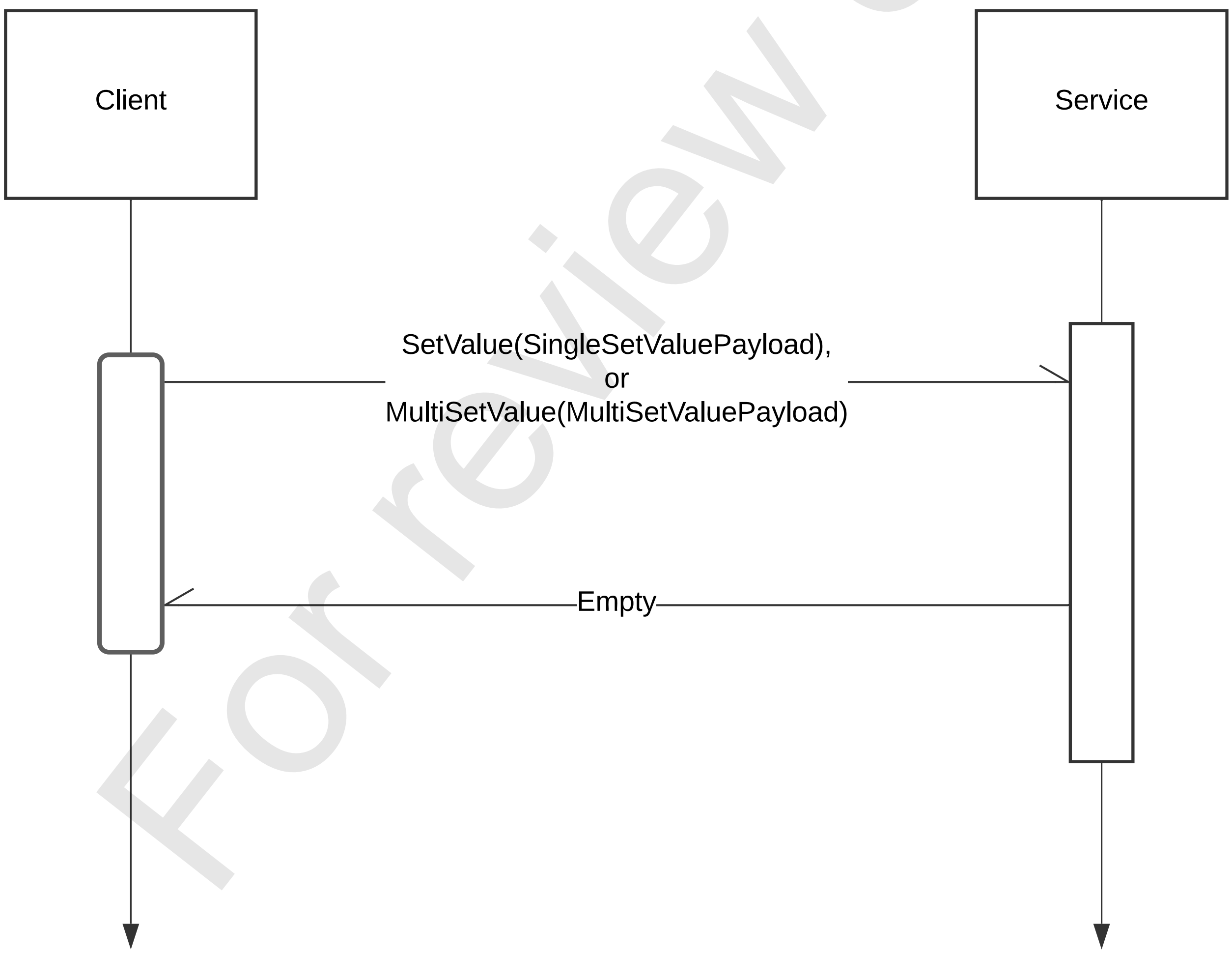

Figure 7 — SingleSetValue and MultiSetValue Sequence Diagram

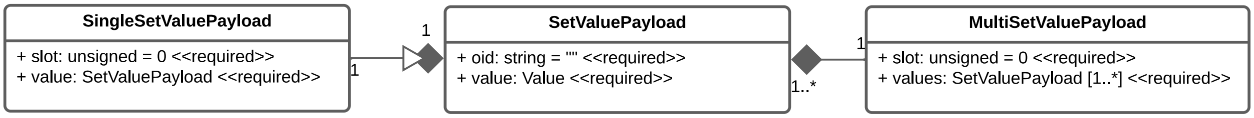

Figure 8 — Set Value Class Diagram

6.3.7 GetValue

Clients use this RPC, illustrated in Table 9 and Figure 9, to query the value of parameters by their object id. The service shall respond with a value if the authenticated user has read access to the parameter's scope.

Table 9 — GetValue RPC

	Transfer Type	Message Definition, file
Request	Unary	GetValuePayload, param.proto
Response	Unary	Value, param.proto
Reference	ST-2138-10 Catena – Model Clause 9.8	

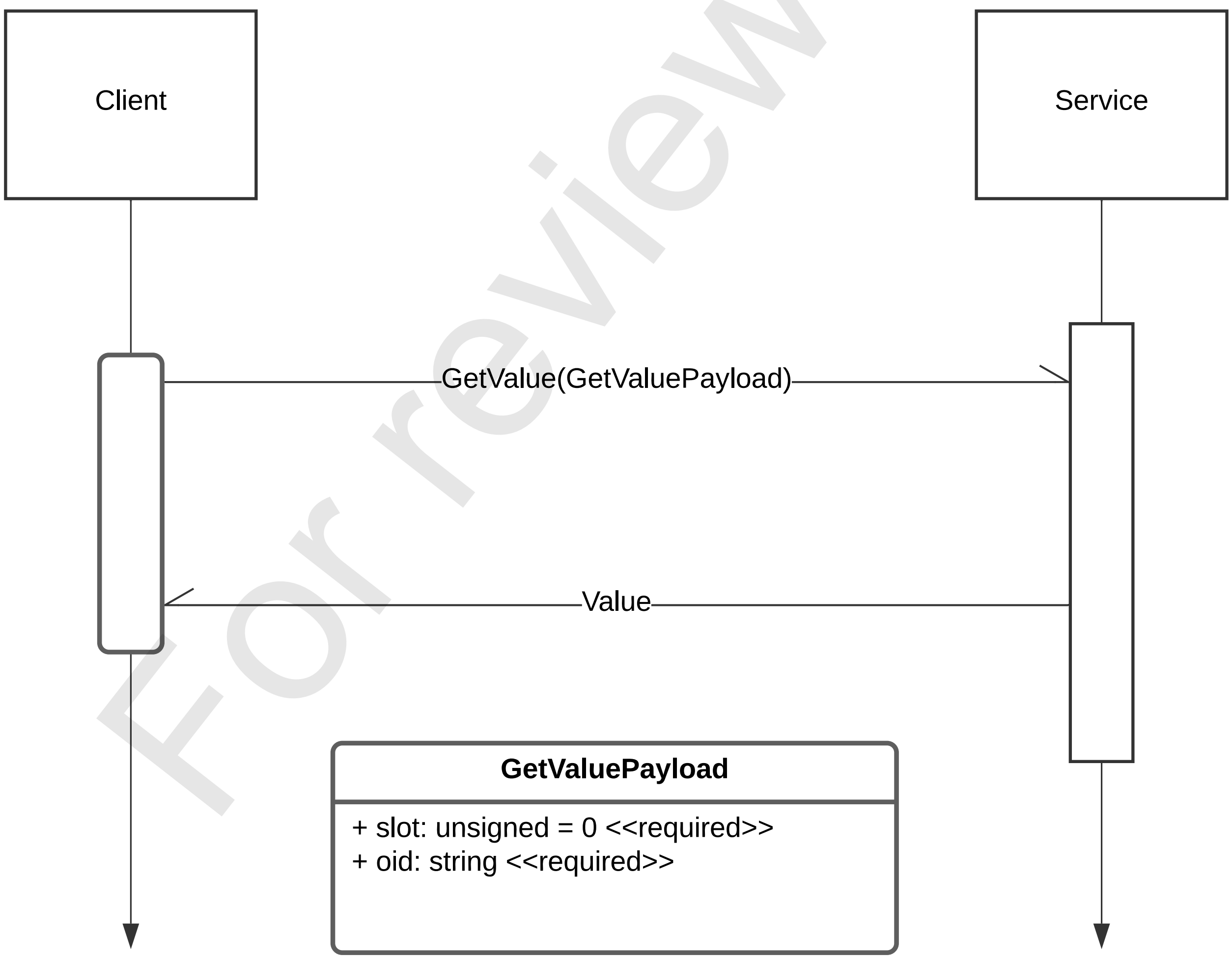

Figure 9 — GetValue RPC

6.3.8 UpdateSubscriptions

This RPC, illustrated in Table 10 and Figure 10, enables a client to change the set of parameters to which it is subscribed. This is useful if the user has switched to a different page of the UI because it allows the client to suppress updates about parameters that are no longer visible, and add the parameters shown on the new page to its subscription.

Table 10 — UpdateSubscriptions RPC

	Transfer Type	**Message Definition, file**
Request	Unary	UpdateSubscriptionsPayload, param.proto
Response	Stream	ComponentParam, device.proto
Reference	SMPTE ST 2138-19:202x Catena – Protocol Objects Clause 6.8	

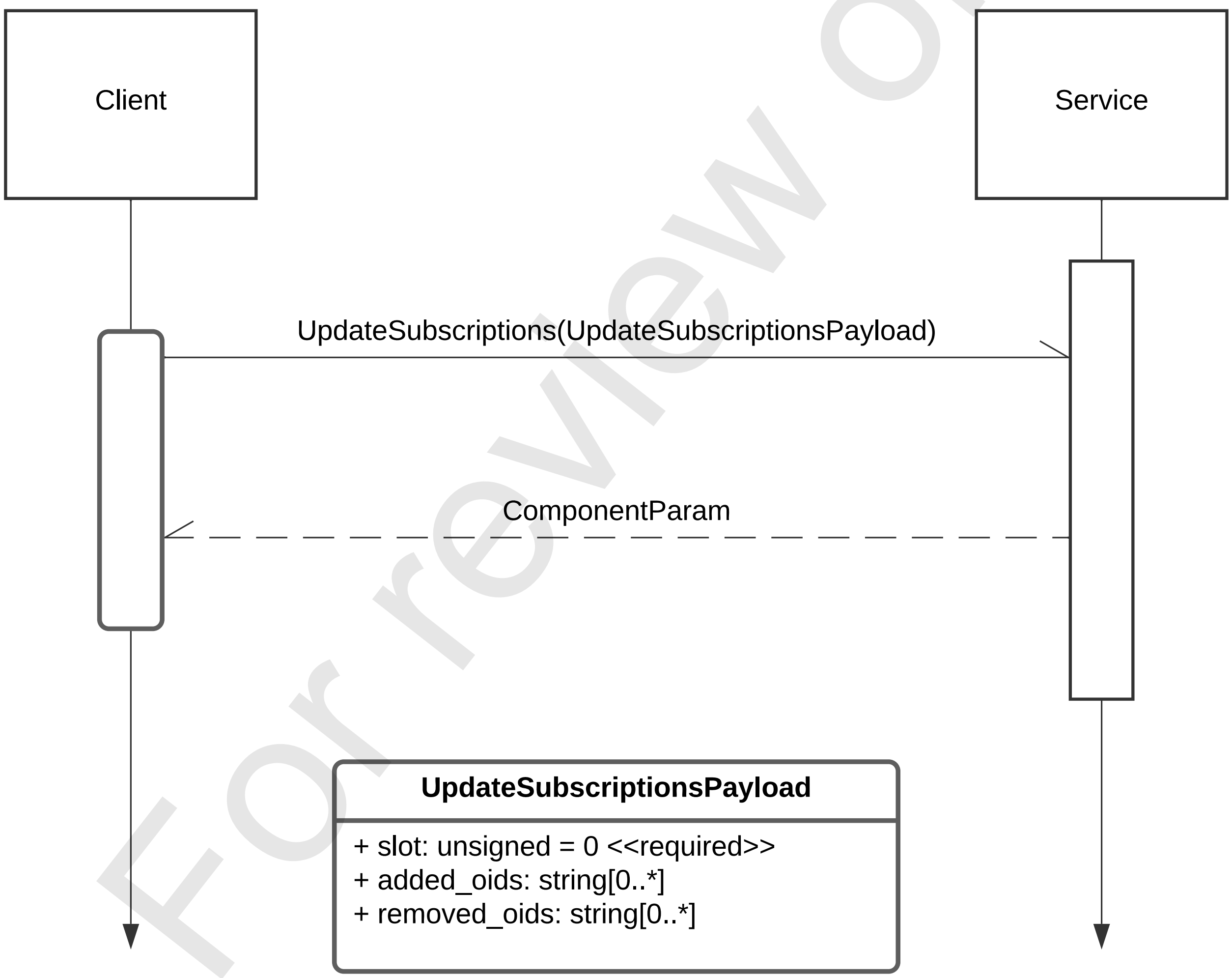

Figure 10 — UpdateSubscriptions RPC

6.3.9 GetParam

This RPC, illustrated in Table 11 and Figure 11, enables a client to query a parameter's descriptor and value. The service shall only provide the requested information if the authenticated user has read access to the parameter's scope. Otherwise, an unauthenticated message shall be returned, as shown in Table 2.

Table 11 — GetParam RPC

	Transfer Type	**Message Definition, file**
Request	Unary	GetParamPayload, param.proto
Response	Unary	ComponentParam, param.proto
Reference	SMPTE ST 2138-19:202x Catena – Protocol Objects Clause 6.8	

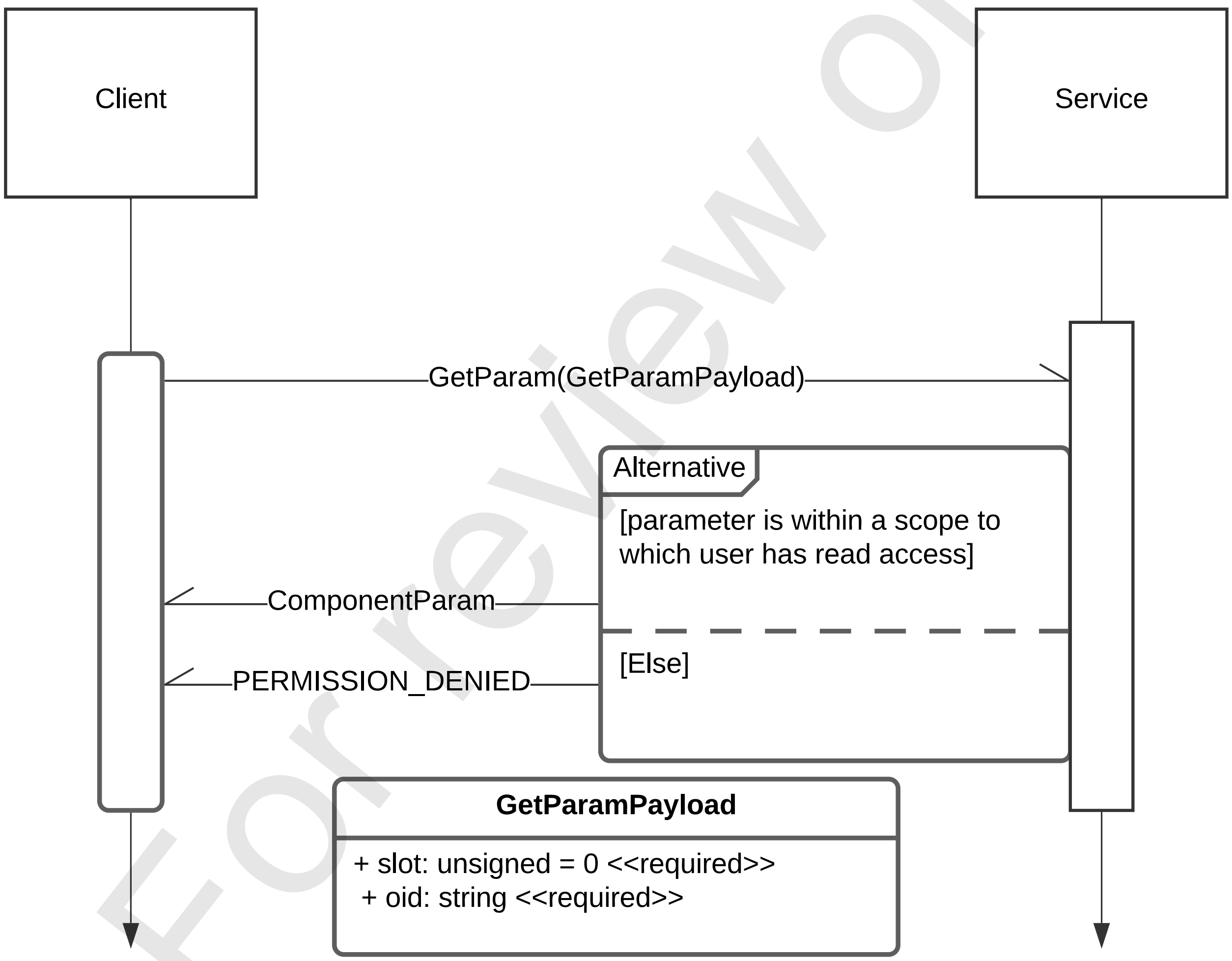

Figure 11 — Get Param RPC

6.3.10 Connect

This RPC, illustrated in Table 12, Figure 12 and Figure 13 connects a client to a service which shall provide push updates until the client cancels the RPC.

Table 12 — Connect RPC

	Transfer Type	Message Definition, file
Request	Unary	ConnectPayload, device.proto
Response	Stream	PushUpdates, device.proto
Reference	SMPTE ST 2138-19:202x Catena – Protocol Objects Clause 6.9	

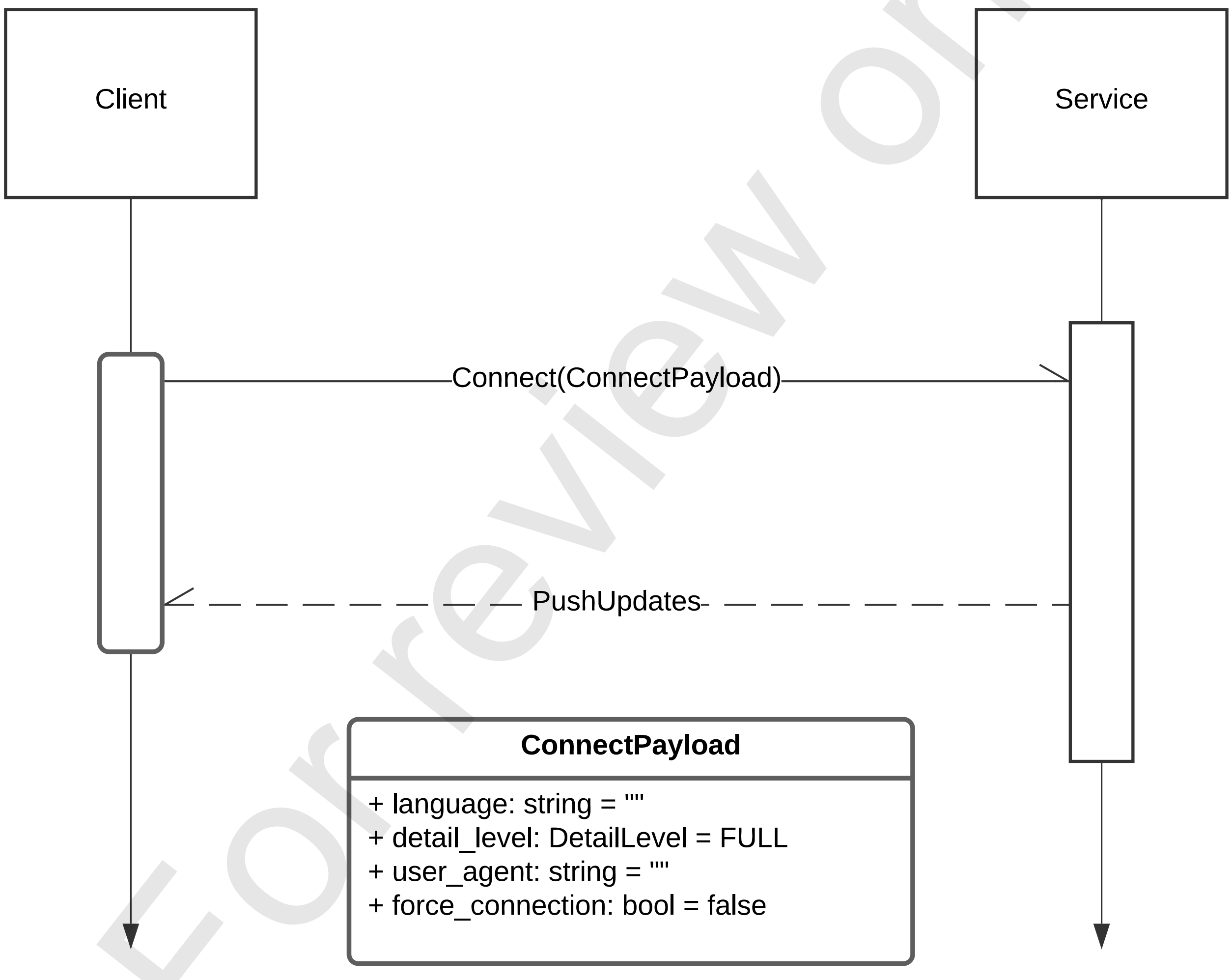

Figure 12 — Connect RPC

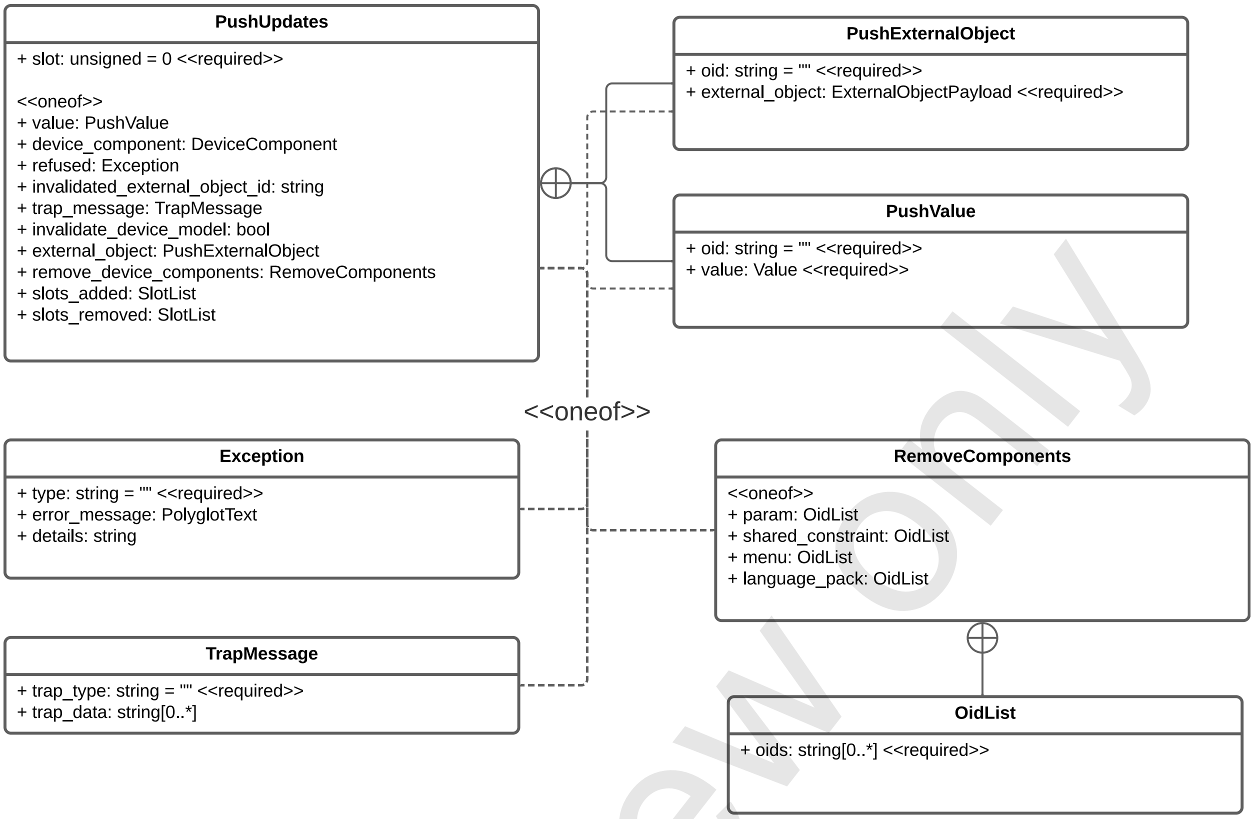

Figure 13 — PushUpdates Class Diagram

6.3.11 RemoveComponents

The RemoveComponents message illustrated in Table 13 has four options that all use the same message type: OidList. The oids listed have one or more of these roots within the device mode, each of which is prepended to their corresponding JSON pointer as specified in Table 13.

Table 13 — Roots for Different RemoveComponents

RemoveComponent <<one of>>	Root within DeviceModel
param	/params
shared_constraint	/constraints
menu	/menu_groups
languages	/language_packs

6.3.12 AddLanguage

This optional RPC, illustrated in Table 14 and Figure 14, enables a client to add language packs to devices in the field.

Table 14 — AddLanguage RPC

	Transfer Type	Message Definition, file
Request	Unary	AddLanguagePayload, language.proto
Response	Unary	Empty
Reference	None in common with REST	

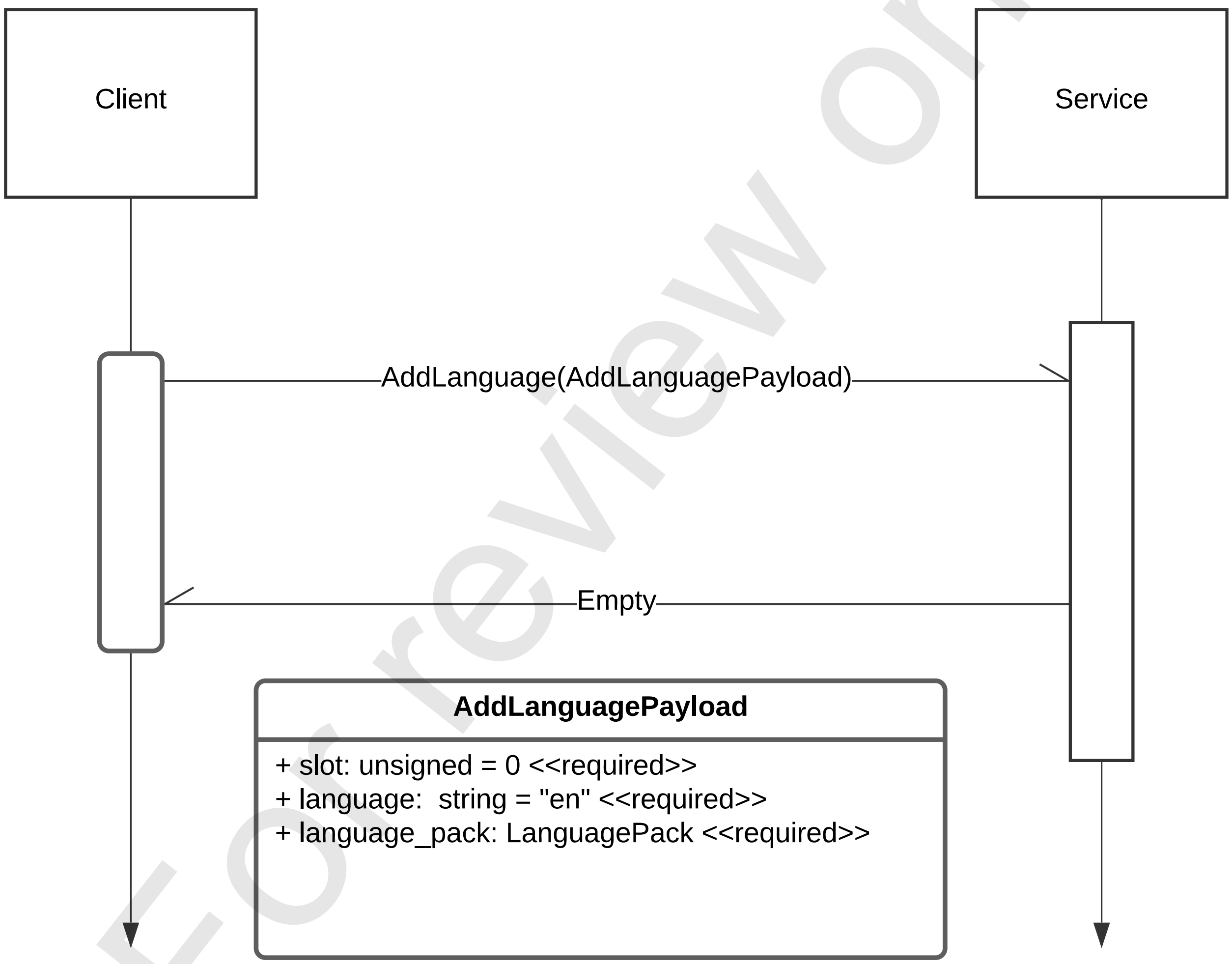

Figure 14 — AddLanguage RPC

6.3.13 LanguagePackRequest

This RPC, illustrated in Table 15 and Figure 15, enables a client to request and receive a language pack by name. The names are registered in IETF RFC 3066 to avoid clashes — e.g., "en", "es", and "fr" for global English, global Spanish, and global French, respectively.

Table 15 — LanguagePackRequest RPC

	Transfer Type	**Message Type, file**
Request	Unary	LanguagePackRequestPayload, language.proto
Response	Unary	ComponentLanguagePack, language.proto
Reference	SMPTE ST 2138-19:202x Catena – Protocol Objects Clause 6.10	

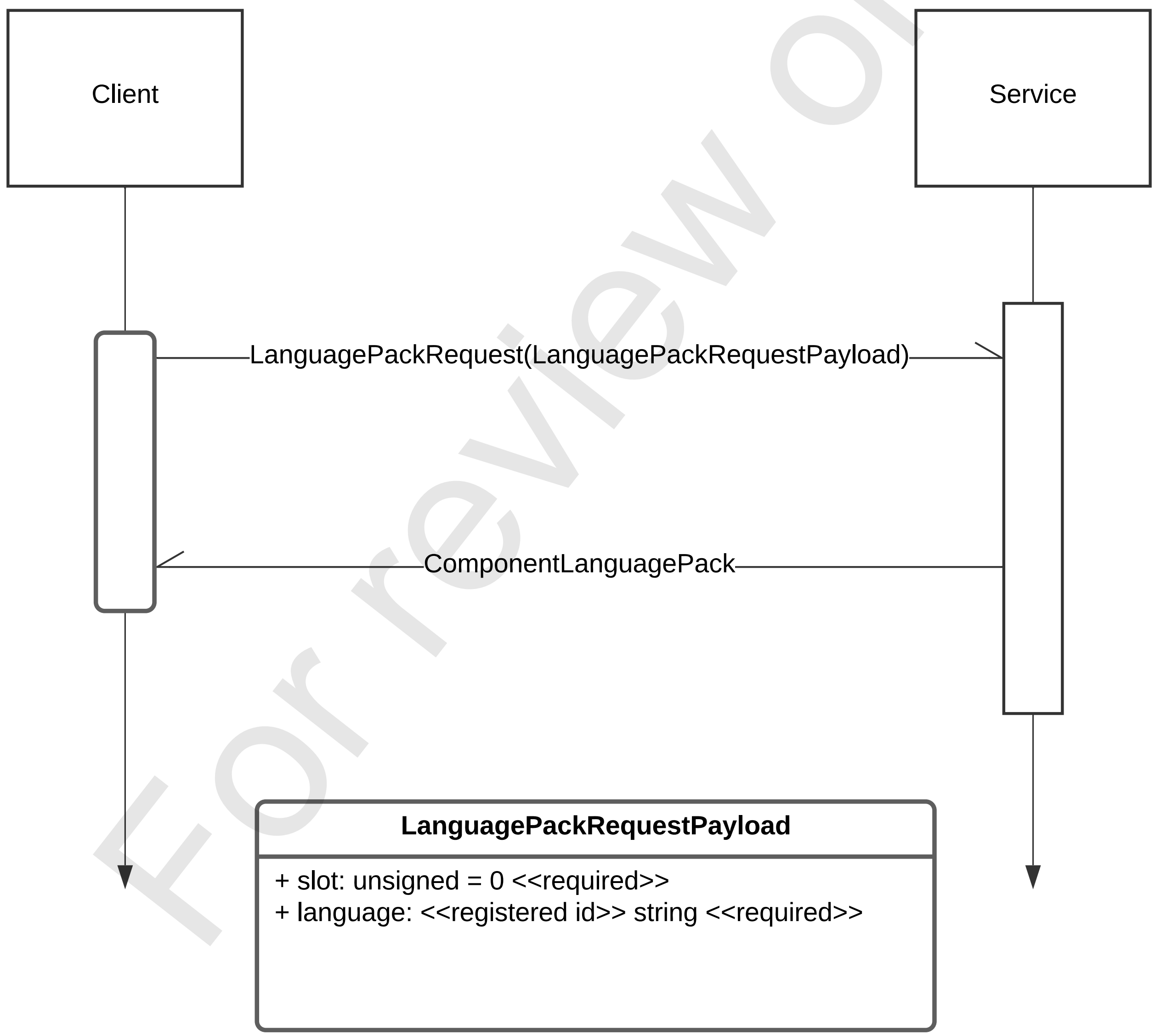

Figure 15 — LanguagePackRequest Class Diagram

6.3.14 ListLanguages

This RPC, illustrated in Table 16 and Figure 16, enables a client to query the languages supported by the device.

Table 16 — ListLanguages RPC

	Transfer Type	Message Type
Request	Unary	ListLanguages
Response	Unary	LanguageList

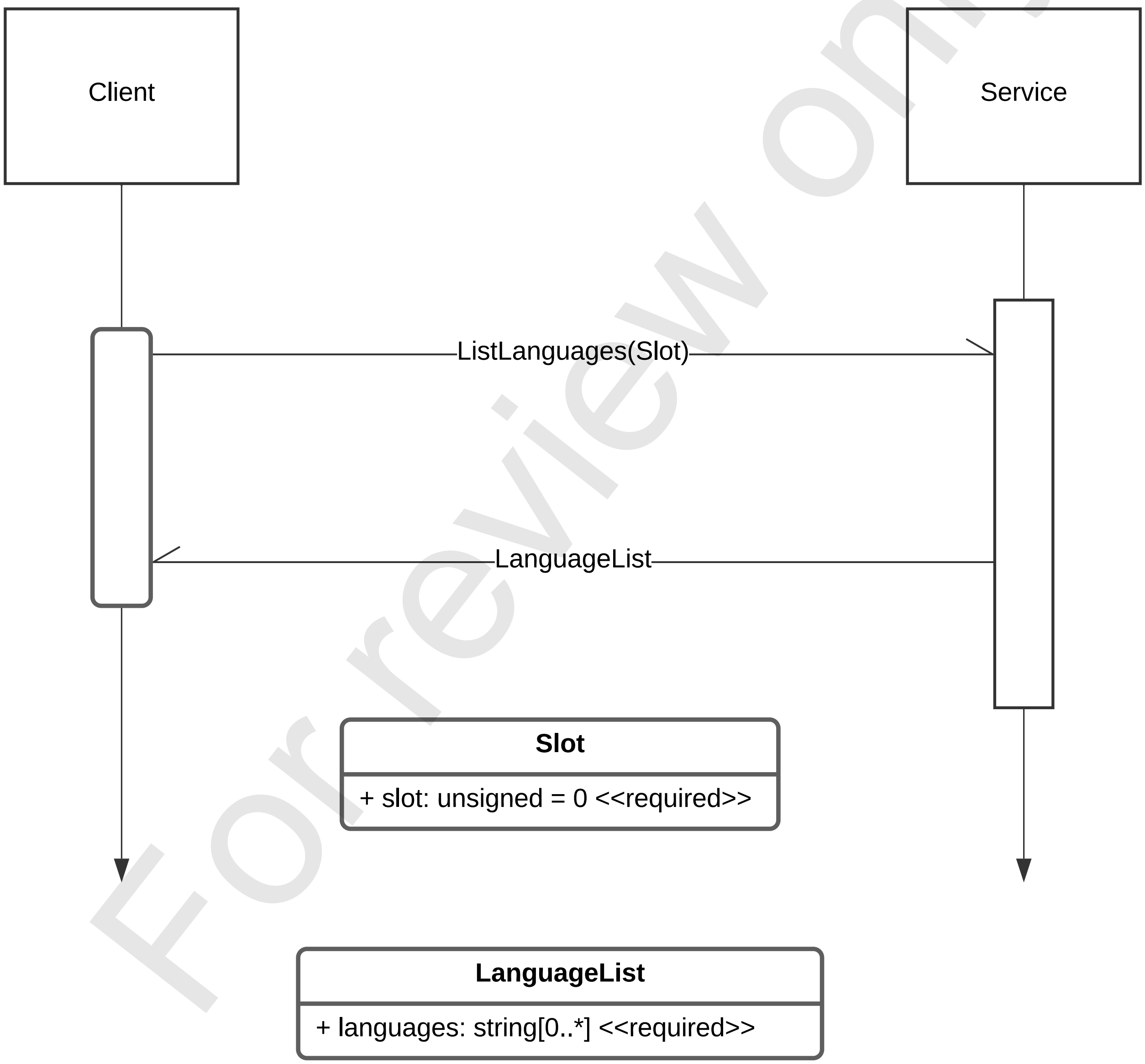

Figure 16 — ListLanguages Sequence and Class Diagram

6.3.15 Access Token Refresh

This RPC, illustrated in Table 17 and Figure 17 attaches a new access token to the metadata which the service is expected to validate and use to extend the expiry of any active streaming RPCs.

Table 17 — RefreshToken RPC

	Transfer Type	Message Type, file
Request	Unary	RefreshTokenPayload, device.proto
Response	Unary	ConnectionStatus, device.proto
Reference	SMPTE ST 2138-19:202x Catena – Protocol Objects Clause 6.11	

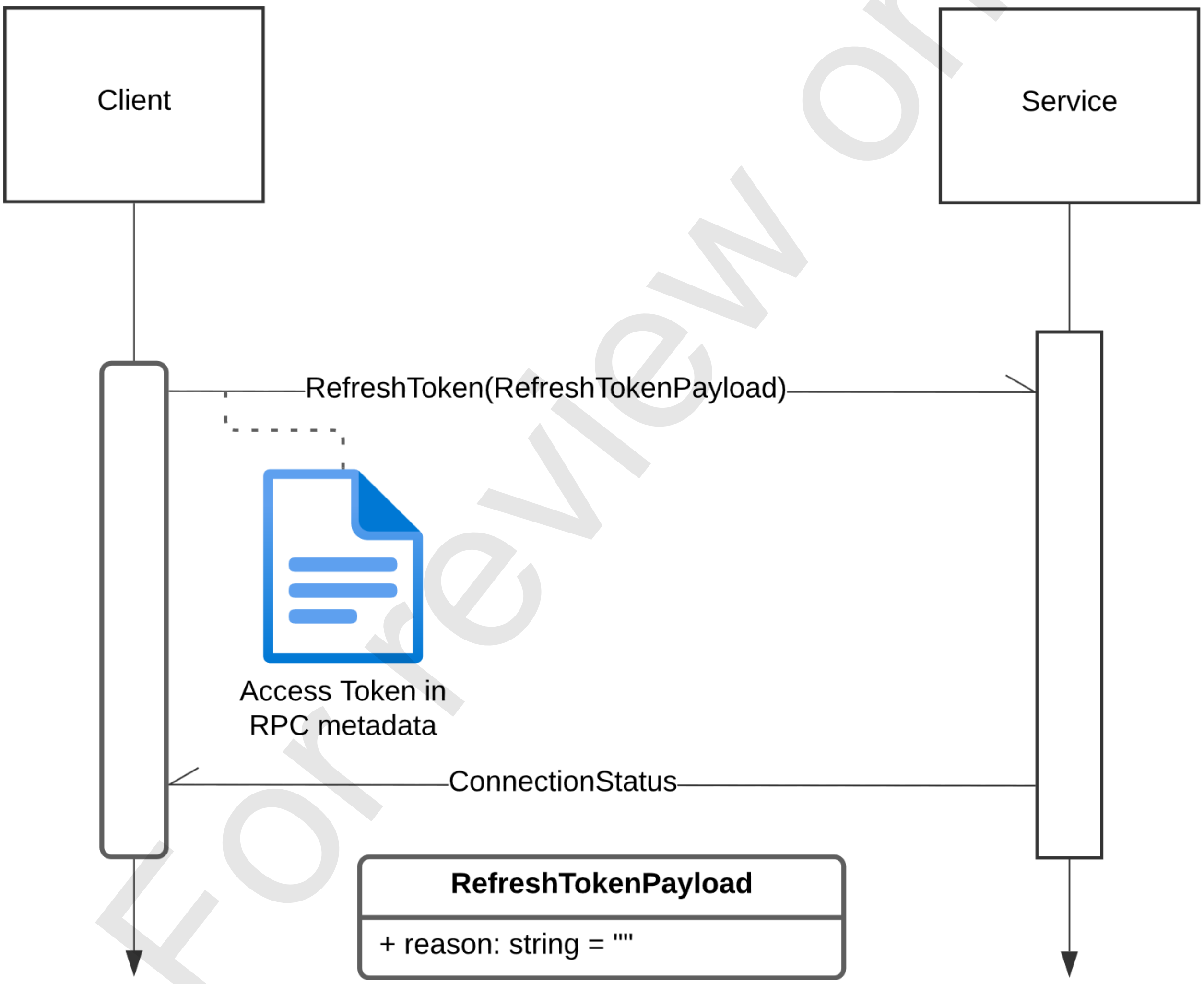

Figure 17 — Refresh Token RPC

6.3.16 Access Revocation

This RPC, illustrated in Table 18 and Figure 18, enables a user with the st2138:adm:w scope in their access token to revoke a subject or all subjects from the device. The response lists the subjects that were revoked and those still connected.

Table 18 — Revoke Access RPC

	Transfer Type	Message Type, file
Request	Unary	RevokeAccessPayload, device.proto
Response	Unary	RevocationResponse, device.proto
Reference	SMPTE ST 2138-19:202x Catena – Protocol Objects Clause 6.12	

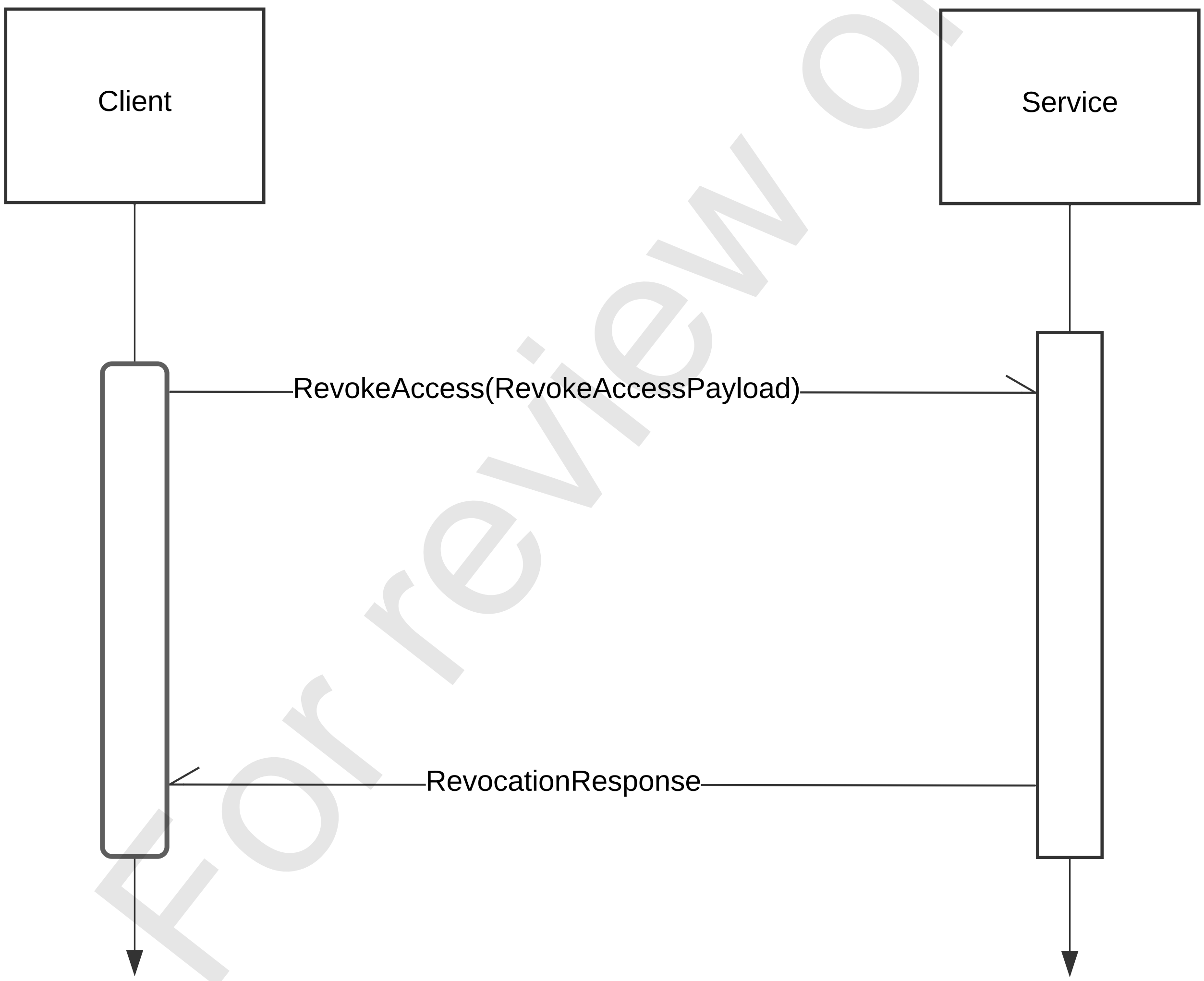

Figure 18 — Revoke Access RPC

Annex A
(Informative)

Additional Elements

A.1 Repository

A set of schemas represented by the UML diagrams in this document provided as a companion element to this document. These can be found in the public repository at the following URL:

https://github.com/SMPTE/st2138-a.git

The contents of this repository are normative.

The software version corresponding to this document revision is tagged as v1.0.0-pcd.

A.2 Root Directory

Throughout this document, the value of the symbol $(ROOT) is the directory into which the test materials are cloned from the repository.

